8. Catalysed reaction

Catalysis is an important branch of surface chemistry which is based on the phenomenon of adsorption of materials on the catalyst surface.

Examples:

In the Haber's process, ammonia is manufactured from N_2 and H_2 as shown by the following reactions.

$$N_2 + 3H_2 \rightarrow 2NH_3$$

In this process, Fe is the catalyst and Mo is a promoter. The surface of the Fe catalyses the reaction.

In the hydrogenation of oils to obtain vanaspathi, Nickel is used as a catalyst. Nickel surface catalyses the reaction.

$$\text{vegetable oil} + H_2 \xrightarrow[473K]{\text{Ni catalyst}} \text{vanaspathi}$$

9. Qualitative analysis

When blue litmus solution is added to Al^{3+} ion, a red coloration is seen due to the acidic nature of the solution. Addition of ammonium hydroxide to it gives a blue lake. This is due to the adsorption of blue colour litmus compound on the surface of $Al(OH)_3$ Which is formed during the addition of NH_4OH

10. Medicine:

Drugs cure diseases by adsorption on body tissues.

11. Concentration of Ores of metals

Sulphides ores are concentrated by a process called froth flotation in which light ore particles are wetted by pine oil.

12. Mordants and Dyes

Most of the dyes are adsorbed on the surface of the fabrics. Mordants are the substances used for fixing dyes onto the fabric.

13. Adsorption indicators

In the precipitation titrations, the end point is indicated by an external indicator which changes its colour after getting adsorbed on precipitate. It is used to indicate the end point of the titration.

10.2 Catalysis

In 1836 Berzelius identified certain substances loosen the bond in the reacting molecules and increased the rate of the reaction. But he also found these substances didn't undergo any change chemically. In order to indicate the property, he gave them the name catalyst. (In greek, kata-wholly, lein-to loosen).

XII U10-Surface Chemistry.indd 76 11-01-2021 15:33:28